



Impacts of the COVID-19 pandemic on financial market connectedness

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ABSTRACT

In this paper, we study the impacts of the COVID-19 pandemic on the connectedness of the Hong Kong financial market. We construct dynamic financial networks based on correlations and partial correlations of stock returns to assess the impacts of COVID-19 and to compare the impacts in the previous financial crises in the past 15 years. Compared to other crises where the network density and clustering can be explained by co-movement with market indices as in normal periods, both network density and clustering are higher in the partial correlation networks during the COVID-19 outbreak.

1. Introduction

The World Health Organization (WHO) declared the coronavirus 2019 (COVID-19) pandemic on 11 March 2020 (WHO, 2020a). As of 9 November 2020, there were around 50 million cases and 1.2 million deaths reported (WHO, 2020b).

This global pandemic also affects the stability of financial markets (Zhang et al., 2020). Ashraf (2020) measured the effect of COVID-19 confirmed cases on stock market return in different countries. Akhtaruzzaman et al. (2020) analyzed the occurrence of financial contagion through both financial and non-financial firms between China and G7 countries. Conlon and McGee (2020) showed that Bitcoin does not serve as a safe haven during the crisis. Corbet et al. (2020) investigated the correlation between stocks in which their names or products are related to “Corona”. Goodell (2020) outlined the social and economic impacts brought by COVID-19 pandemic. Haroon and Rizvi (2020) conducted sentiment analysis on news related to COVID-19 and linked it to volatility of equity markets. Zhang et al. (2020) studied the country-specific risks and systemic risks among global markets. Baig et al. (2020) studied the impact of the COVID-19 pandemic, in terms of the liquidity and volatility, on the equity markets in the US. Heyden and Heyden (2020) investigated the market reactions of the US and European stocks at an early stage of the COVID-19 pandemic. Cepoi (2020) provided evidence on the impact of the COVID-19 pandemic to the returns of the stock market from top six countries which are mostly affected. Corbet et al. (2020) suggested that neither gold nor Bitcoin is significantly correlated with Chinese stock markets during the pandemic. Dwita Mariana et al. (2020) studied whether the two largest cryptocurrencies, Bitcoin and Ethereum, can be a safe-havens for stocks during the COVID-19 pandemic. Goodell and Goutte (2020) used wavelet methods to study the relationship between daily COVID-19 world deaths and daily Bitcoin prices. Shehzad et al. (2020) compared the impact of the COVID-19 pandemic on the variance of markets in several countries around the globe with the impact of global financial crises. So et al. (2020); Chu et al. (2020a, 2020b) used

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network analysis to construct pandemic networks for understanding the propagation of the COVID-19 pandemic risk.

In this paper, we use network analysis to study the impacts of the COVID-19 pandemic on the connectedness of stock returns in the Hong Kong financial market by constructing dynamic financial networks. The remaining of the paper is as follows. Section 2 describes the statistical methods we used to construct dynamic financial networks and to conduct network analysis. Section 3 gives the results of our analysis, and Section 4 concludes.

2. Network analysis

2.1. Dynamic network construction

Define the daily return at time t for stock i by $r_i(t) = \log(p_i(t)) - \log(p_i(t-1))$, where $p_i(t)$ is the adjusted closing price of stock i at time t , and $i = 1, \dots, n$. Using the financial returns, we construct dynamic financial networks based on a moving-window approach. We estimate the correlation between stock i and stock j at time t by calculating the sample correlation, $\hat{\rho}_{ij}(t)$, using the recent k -day of information, i.e. from day $t-k+1$ to day t . The correlation $\hat{\rho}_{ij}(t)$ is statistically significant at α level if

$$\left| \hat{\rho}_{ij}(t) \sqrt{\frac{k-2}{1-\hat{\rho}_{ij}(t)^2}} \right| > t_{k-2}(1-\alpha/2), \quad (1)$$

where t_{df} is the quantile function of t distribution with degrees of freedom df . With the p-values from the individual hypothesis tests, we also control the family-wise error rate by applying Holm's correction (Holm, 1979). In the following discussion, we present two sets of results; one is based on the network construction using individual hypothesis tests and the other is based on the network construction using the hypothesis tests with Holm's correction.

We construct a time series of networks $G(t) = (V(t), E(t), W(t))$ to show the relationship among stocks at time t . Let $V(t)$ to be the set of all vertices, referring to the stocks being considered in the network at time t . If some stocks are not yet listed on the market at time t , we do not include them to the set $V(t)$. Let $E(t)$ be the set of edges, consisting of distinct pair of stocks satisfying some properties of interest. In this study, we define the edge between stock i and stock j at time t when (1) holds. Finally, we let $W(t)$ be an $n \times n$ matrix storing the weight of edge. If two distinct stocks i and j are connected at time t , the (i, j) th entry of $W(t)$ is the correlation of returns between stock i and j . Otherwise, the (i, j) th entry of $W(t)$ is 0.

To separate the effect of market factors from the co-movement of returns of individual stocks, we also study correlations between idiosyncratic returns. Let $r_M(t)$ be the market factor return at time t . Specifically, we assume the following market model

$$r_i(t) = a_i + b_i^\top r_M(t) + \varepsilon_i(t),$$

where $\varepsilon_i(t)$ is the error term with mean zero and constant variance. The idiosyncratic returns are defined as $r_{i|M}(t) = r_i(t) - \hat{r}_i(t)$, where $\hat{r}_i(t) = \hat{a}_i + \hat{b}_i^\top r_M(t)$ is the estimated return based on the fitted market model using the data from time $t-k+1$ to time t , and $\hat{a}_i$ and $\hat{b}_i$ are estimates of a_i and b_i . Using the idiosyncratic returns, we can calculate $\hat{\rho}_{ij|M}(t)$, the sample partial correlation between stock i and stock j at time t . In the financial networks based on partial correlation, we connect stock i and stock j if $\hat{\rho}_{ij|M}(t)$ is significant at α level, or when

$$\left| \hat{\rho}_{ij|M}(t) \sqrt{\frac{k-m-2}{1-\hat{\rho}_{ij|M}(t)^2}} \right| > t_{k-m-2}(1-\alpha/2), \quad (2)$$

where m is the dimension of $r_M(t)$. The dynamic network on partial correlations captures the co-movement of idiosyncratic returns which cannot be explained by common market factors. Exploring these dynamic networks helps us to understand the evolution of systemic risk.

To assess the impact on the correlations and partial correlations during the crisis periods, we compare the results where the correlations and partial correlations are calculated with the adjustment for volatility using the method in Forbes and Rigobon, 2002. We find that the empirical results are very similar to those with the volatility adjustment. Therefore, we keep the unadjusted correlations, $\hat{\rho}_{ij}(t)$ and $\hat{\rho}_{ij|M}(t)$, and their results throughout the discussion. The empirical results with the volatility adjustment are presented in the Supplementary Information.

2.2. Summary statistics of networks

In our study, we focus on those edges with positive correlations (or partial correlations) to calculate three network statistics. For a fixed significant level α in (1) and (2), let $G_\alpha(t) = (V_\alpha(t), E_\alpha(t), W_\alpha(t))$ be the network at time t with stocks being connected if their correlation or partial correlation is significant at α level. The first statistic is the average weight of $G_\alpha(t)$, which is defined by

$$\bar{W}_\alpha(t) = \frac{2}{n(n-1)} \sum_{i < j} W_{ij}(t).$$

where $|E_\alpha(t)|$ is the number of edges in the network $G_\alpha(t)$.

Average local clustering coefficient $C(t)$, our second network statistic, is defined by

$$C(t) = \frac{1}{n} \sum_{i=1}^n C_i(t),$$

where $C_i(t)$ is the local clustering coefficient for node i , the ratio between the number of edges and all possible edges in the subgraph of $G_\alpha(t)$ containing all the nodes which have an edge joining node i , that is

$$C_i(t) = \frac{|\{\{j, k\} \in E_\alpha(t) : \{i, j\}, \{i, k\} \in E_\alpha(t), i, j, k \in V_\alpha(t)\}|}{|\{\{j, k\} : \{i, j\}, \{i, k\} \in E_\alpha(t), i, j, k \in V_\alpha(t), j \neq k\}|}. \quad (3)$$

In the cases when node i has a degree equal to 0 or 1, the denominator in (3) is 0. In those cases, we set $C_i(t) = 0$ to reflect the weak influential power of node i to the connectedness of the network.

The average local clustering coefficient, $C(t)$, measures the tendency of two nodes to be linked at time t if they are linked with one another common node. If the value is large, for correlation networks, we shall expect that the diversification effect of a portfolio formed by the stocks in the network will not be strong, as the stocks are highly connected. When the clustering coefficient in a partial correlation network is large, we shall expect that the co-movement of stocks in the network cannot be effectively explained by common market factors, and thus the hedging effectiveness due to the market factors for a portfolio formed by the stocks in the network will not be high.

Network density $ND(t)$, our third statistic, is the ratio between the number of edges, $|E_\alpha(t)|$, and the number of all possible edges in the network at time t , i.e.

$$ND(t) = \frac{|E_\alpha(t)|}{|V_\alpha(t)|(|V_\alpha(t)| - 1)/2},$$

where $|V_\alpha(t)|$ is the number of nodes in the network $G_\alpha(t)$.

3. Empirical study

To demonstrate the dynamic network approach, we analyze $n = 100$ stocks with the highest market capitalization listed in the Stock Exchange of Hong Kong from 3 Jan 2006 to 30 Oct 2020, with a total $T = 3663$ trading days. We choose $m = 2$ market indices, the Hang Seng Index (HSI) and the Hang Seng China Enterprises Index (HSCEI) to build the partial correlation networks. We choose the width of those moving windows as $k = 40$ trading days because we want to include sufficient number of observations to make a conclusion. We have also repeated the empirical study with different window widths, but we found that the fluctuation of the statistics mentioned in the above section tends to be large when we are considering a small window width. We highlight the interval from Oct 2007 to Mar 2009, Apr 2011 to Aug 2012 (Xu et al., 2017), Jun 2015 to early Feb 2016 (Khojine and Han, 2019), and Jan 2020 onward (Zhang et al., 2020) to indicate four major financial crises in Hong Kong during the period of study. These four periods refer to the subprime mortgage crisis, European debt crisis, Chinese stock market turbulence, and the COVID-19 pandemic.

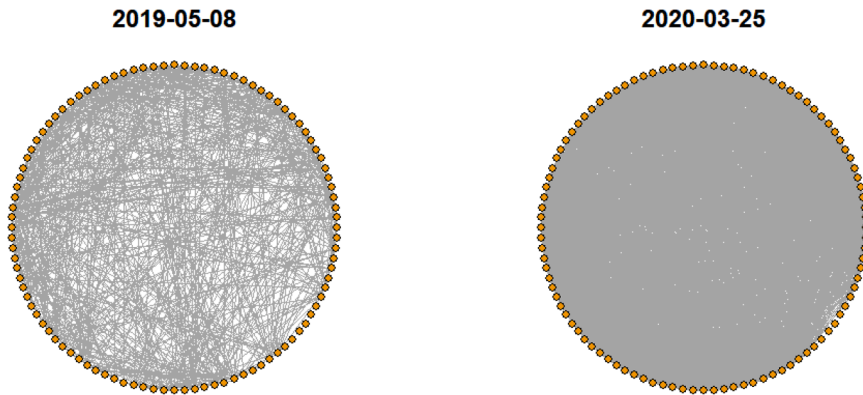


Fig. 1. The network constructed based on the partial correlation at 5% significant level on 8 May 2019 and 25 Mar 2020, before and during the COVID-19 pandemic, respectively.

3.1. Network visualization

Fig. 1 shows the network based on the partial correlation at 5% significance level on 8 May 2019 and 25 Mar 2020. We choose these two days because the average partial correlations, among those significant at 5%, are the highest before and during COVID-19 outbreak respectively. It is obvious that the second plot is more dense, indicating that more edges exist during the COVID-19 outbreak than those before the COVID-19 outbreak. The visual display in Fig. 1 presents much higher network connectedness during the outbreak. Fig. 2 shows the corresponding networks at the days with the highest partial correlation at 5% level with Holm's correction before and during COVID-19 outbreak. With Holm's correction, there are fewer significant correlations and so the network maps in Fig. 2 are less dense than that in Fig. 1. However, the message from the two figures is the same: stocks are much more connected during COVID-19 outbreak than other trading days in the past 15 years.

3.2. Average weight

Fig. 3 presents the average correlation, $\bar{W}_\alpha(t)$, for $\alpha = 5\%$, 1% and 0.2% . For $\alpha = 5\%$, $\bar{W}_\alpha(t)$ increases from 0.1 in normal period to a peak at 0.15 during the first financial crisis, from 0.15 to 0.3 during the second and third. During COVID-19 outbreak, the average correlation attains the highest value at around 0.5. A smaller significance level, like 1% and 0.2% , also shows a similar pattern to that for $\alpha = 5\%$. With Holm's correction to construct networks, $\bar{W}_\alpha(t)$ is smaller because there are fewer edges in the networks. Fig. 4 shows that with Holm's correction, we can still see a sharp increase in the average weight. For $\alpha = 5\%$, $\bar{W}_\alpha(t)$ increases from 0.05 in normal period to a peak of 0.1 during the first financial crisis, from 0.05 to 0.2 during the second and third crises. During COVID-19 outbreak, the average correlation attains the highest value at around 0.4.

For the partial correlation networks with $\alpha = 5\%$, $\bar{W}_\alpha(t)$ stays at an interval between 0 to 0.05 before COVID-19 outbreak. However, it jumps to the highest value at around 0.25 during COVID-19 outbreak. As in the plot for correlation, the pattern of partial correlation curve corresponding to $\alpha = 1\%$ and 0.2% are similar to the one of 5% . With Holm's correction, the increase in the partial correlation plot during COVID-19 outbreak is still sharp. The average weight $\bar{W}_\alpha(t)$ stays at an interval between 0 to 0.005 before COVID-19 outbreak. However, it jumps to the highest value at around 0.05 during COVID-19 outbreak.

The behavior of the partial correlation networks during the outbreak is very different from that in the first three financial crises. Stocks in financial markets tend to have common downward movement during financial turmoil. In the first three financial crises, although there are additional significant and strong correlations, they are well-explained by the market factors. However, during COVID-19 outbreak, the market factors cannot explain adequately the additional individual co-movement of stocks, leading to higher partial correlations during COVID-19 outbreak than before the outbreak.

3.3. Clustering coefficient and network density

Fig. 5 gives the average local clustering coefficient, $C(t)$, of the correlation and partial correlation networks of those significant at 5% , 1% and 0.2% levels. The correlation networks has $C(t)$ reaching around a peak at around 0.5, 0.7, 0.75, and 0.85 in the four financial crises respectively, which are higher than the normal period right before the crisis. If we use a smaller significance level, the increasing trend during crises remains. In addition, Fig. 6 shows that with Holm's correction, the pattern in the average local clustering coefficient is similar. The correlation networks have $C(t)$ reaching a peak at around 0.35 in the first, around 0.6 in the second and third crises, and around 0.75 during COVID-19 outbreak.

In the partial correlation networks of those significant at 5% , 1% and 0.2% levels, the average local clustering coefficients vary between 0.2 and 0.35 in most of the time before COVID-19 outbreak, but show a sharp acceleration to the peak at around 0.75 during COVID-19 outbreak. Again, the above observations remain valid with different values of α level. With Holm's correction, the average

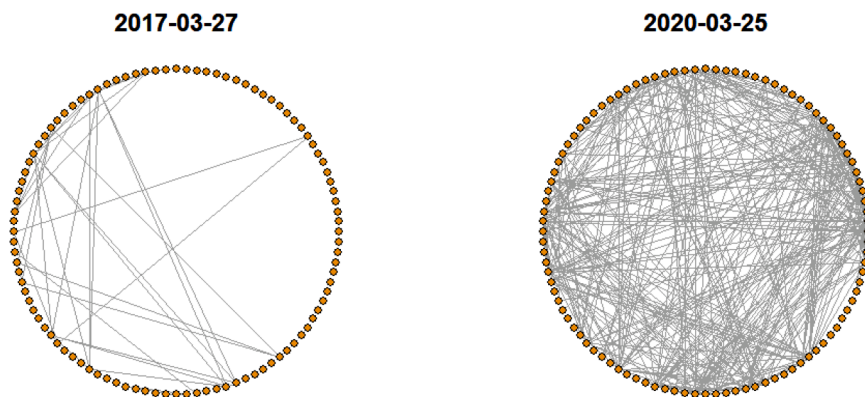


Fig. 2. The network constructed based on the partial correlation at 5% significant level with Holm's correction on 27 Mar 2017 and 25 Mar 2020, before and during the COVID-19 pandemic, respectively.

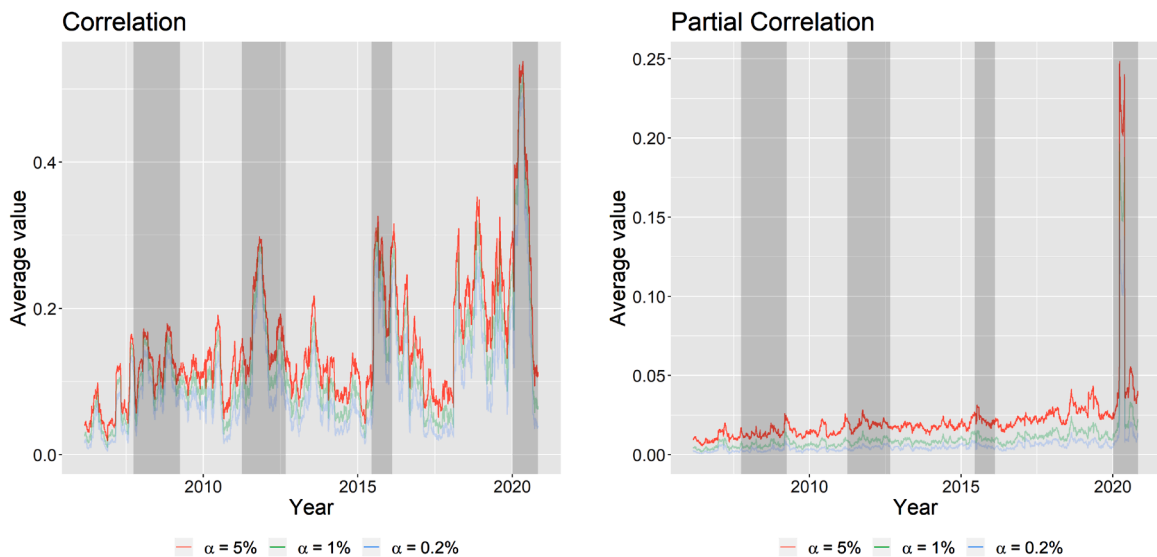


Fig. 3. The average correlation (left) and average partial correlation (right) of those stock pairs at 5%, 1% and 0.2% significance levels.

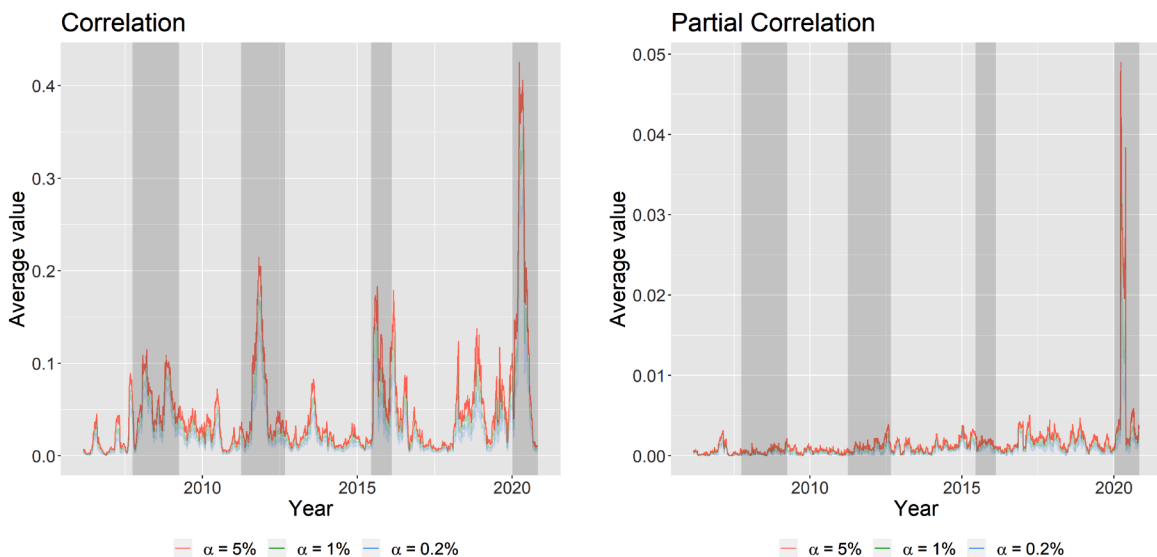


Fig. 4. The average correlation (left) and average partial correlation (right) of those stock pairs at 5%, 1% and 0.2% significance levels with Holm's adjustment.

local clustering coefficients vary between 0.05 and 0.1 in most of the time before COVID-19 outbreak, but show a sharp acceleration to the peak at around 0.3 during COVID-19 outbreak.

Fig. 7 presents the network density, $ND(t)$, of the correlation and partial correlation networks at 5%, 1% and 0.2% levels. The network density in the correlation networks increases to around 0.8 during COVID-19 outbreak. The density is higher in the crisis periods than in normal periods. Similar patterns can be found with different significance levels. Considering the plots with Holm's correction in Fig. 8, the network density in the correlation networks increases to around 0.6 during COVID-19 outbreak.

In the partial correlation networks, the network density is relatively stable at around 0.05 to 0.1 in the first three crises. However, there is a sudden big jump to 0.5 in the network density of the partial correlation networks during COVID-19 outbreak, signifying sharp increase in network connectedness. We can see a similar pattern with different significant levels. The plot corresponding to the network density of the partial correlation networks with Holm's correction shows that there is a jump to 0.07 during COVID-19 outbreak from normal values of around 0.005.

Although in both correlation and partial correlation networks, we find an increase in the network connectedness during the COVID-19 outbreak, the evidence is stronger in partial correlations. During adverse movement in financial markets, the diversification effect is weaker than normal periods due to the co-movement of stocks triggered by extraordinary events. In the first three crises, we can largely

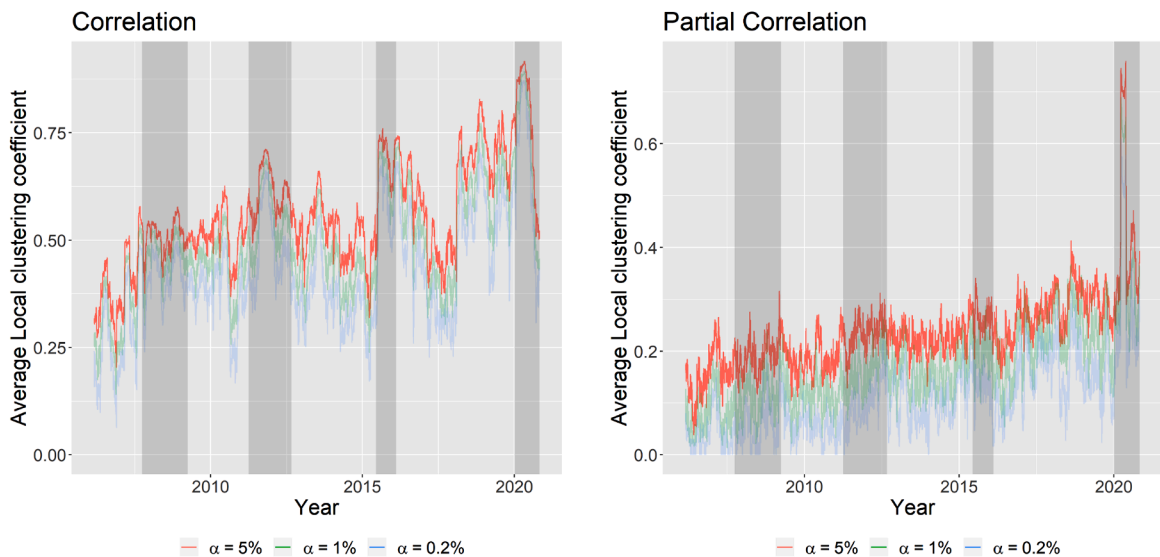


Fig. 5. The average local clustering coefficient of networks corresponding to correlation (left) and partial correlation (right) at 5%, 1% and 0.2% significance levels.

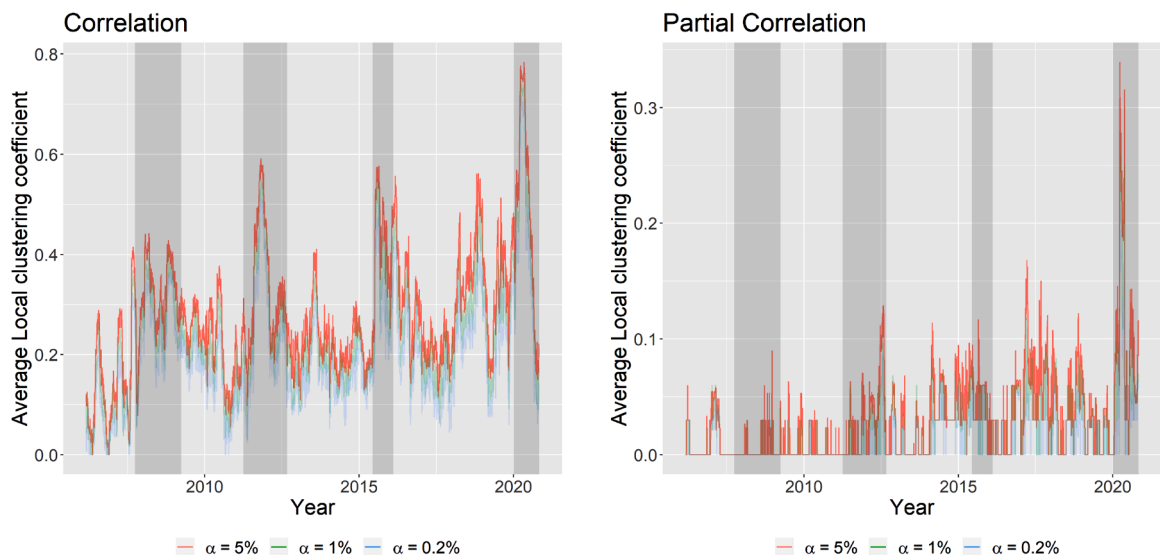


Fig. 6. The average local clustering coefficient of networks corresponding to correlation (left) and partial correlation (right) at 5%, 1% and 0.2% significance levels with Holm's adjustment.

explain the co-movement of stock returns by the market factors. However, during the COVID-19 pandemic, the unusual clustering coefficient and network density give a sign of increasing difficulty to hedge, as the co-movement among stock returns is still strong after controlling the effect of the market factors. Overall, the high network connectedness implies the substantial increase in systemic risk during the outbreak.

4. Conclusion

In this paper, we compare the connectedness of stock returns in four recent financial crises. Contrast to the first three crises, the market factors cannot explain adequately the co-movement of stock returns during the COVID-19 outbreak, leading to a substantial increase in the network connectedness in the financial networks in March and April 2020. The large increase in the connectedness of the partial correlation networks implies much increase in systemic risk in the financial system during the COVID-19 outbreak. A potential reason of the above findings, as suggested by [Pukthuanthong and Roll \(2009\)](#), is that the stock returns are integrated to the market factors, and in general having a small stock-specific volatility. As such, the unexpected downside risk may not be adequately

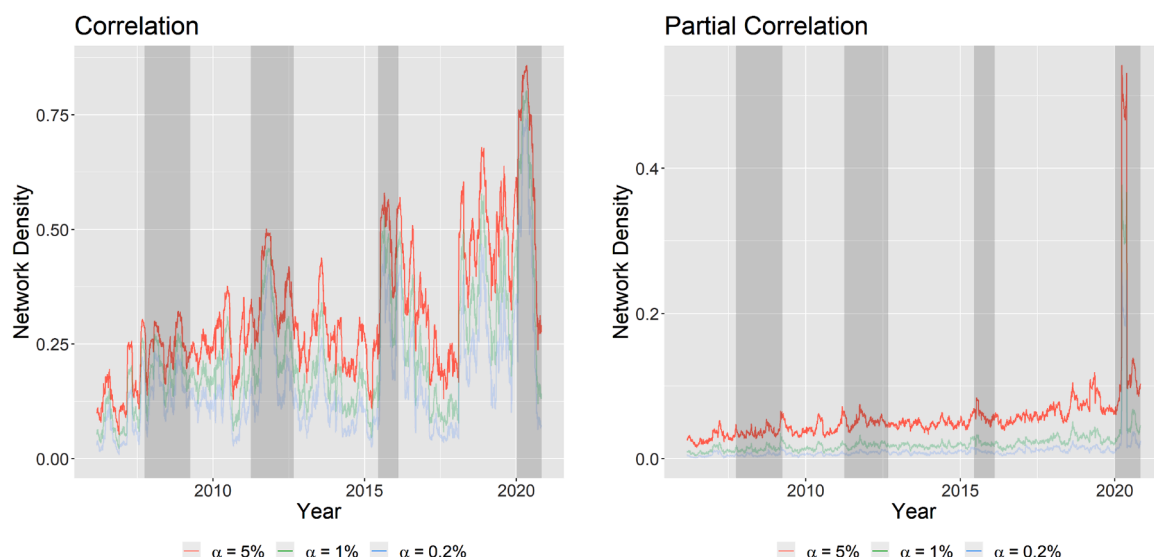


Fig. 7. The network density of networks corresponding to correlation (left) and partial correlation (right) at 5%, 1% and 0.2% significance levels.

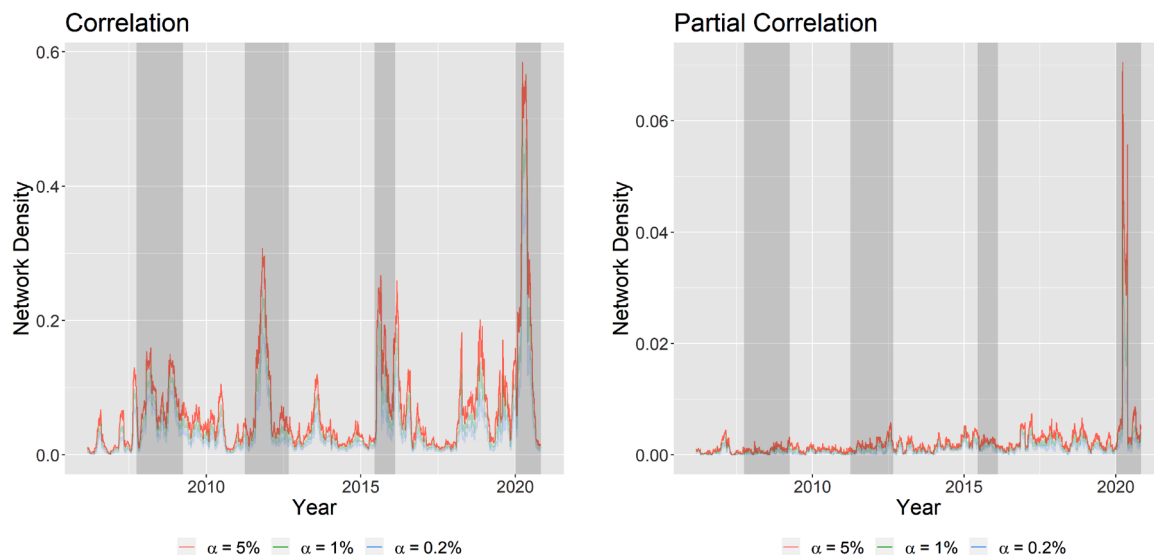


Fig. 8. The network density of networks corresponding to correlation (left) and partial correlation (right) at 5%, 1% and 0.2% significance levels with Holm's adjustment.

explained by the market factors during financial crisis.

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CRediT authorship contribution statement

Mike K.P. So: Conceptualization, Methodology, Writing - review & editing, Supervision. **Amanda M.Y. Chu:** Methodology, Validation, Writing - original draft. **Thomas W.C. Chan:** Data curation, Writing - original draft, Visualization.

Declaration of Competing Interest

All authors have no conflicts of interest.

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Supplementary material

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